MOS Capacitor

- Introduction
- Ideal MOS Capacitor
  - Qualitative Description of MOS Capacitor (CAP)
  - Quantitative Analysis of MOS Capacitor (CAP)
- *C-V* Curves of MOS Capacitor
- Non Ideal MOS Capacitor
- Applications of *C-V* Curve

Quantitative Analysis

- The goal of quantitative analysis is to analyze for charge distribution $\rho(x)$, electric field $\xi(x)$, and potential $\Phi(x)$ as a function of V_G, N_A, T_{ox}

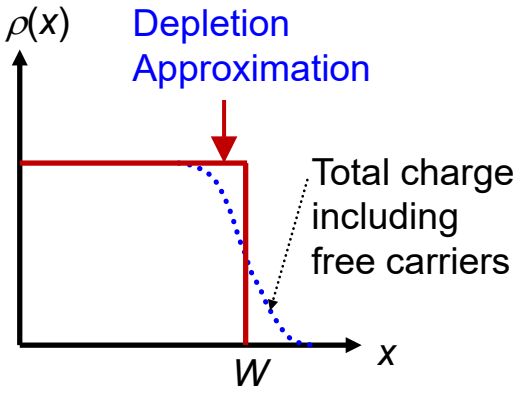
- Following Two Approaches:
  - Depletion Approximation
  - General Analysis

Depletion Approximation

1. n~0, p~0 in the depletion region; mobile charges in the depletion region are totally absent.

2. $p << N_A$ within W $\quad \rho = -q \cdot N_A$

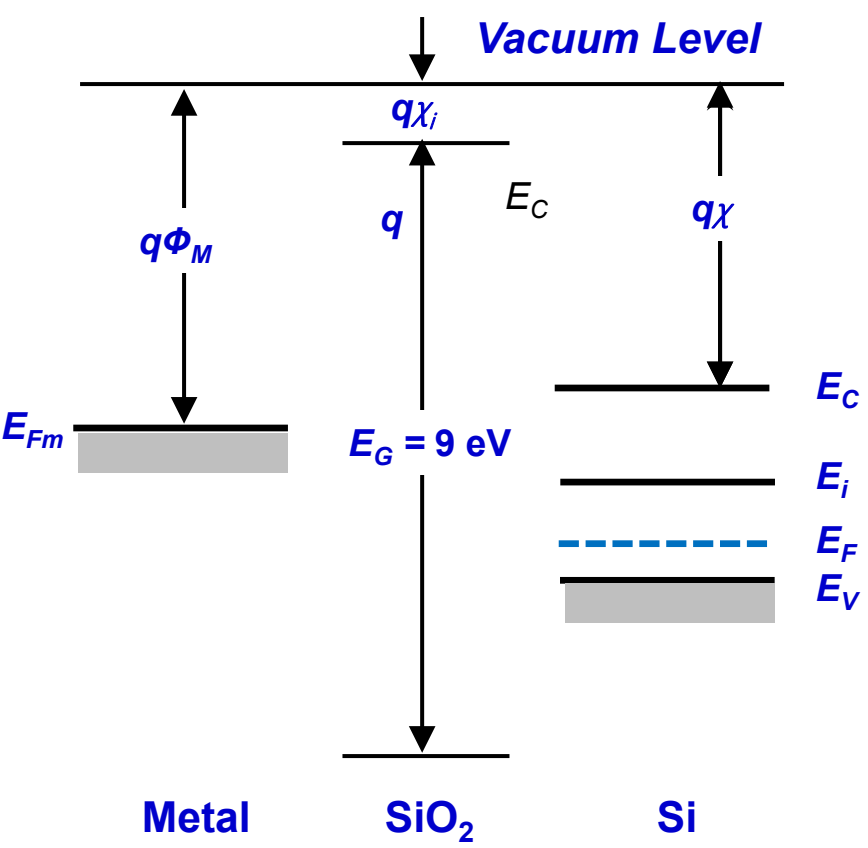
   $p = N_A$ outside W $\quad \rho = 0$

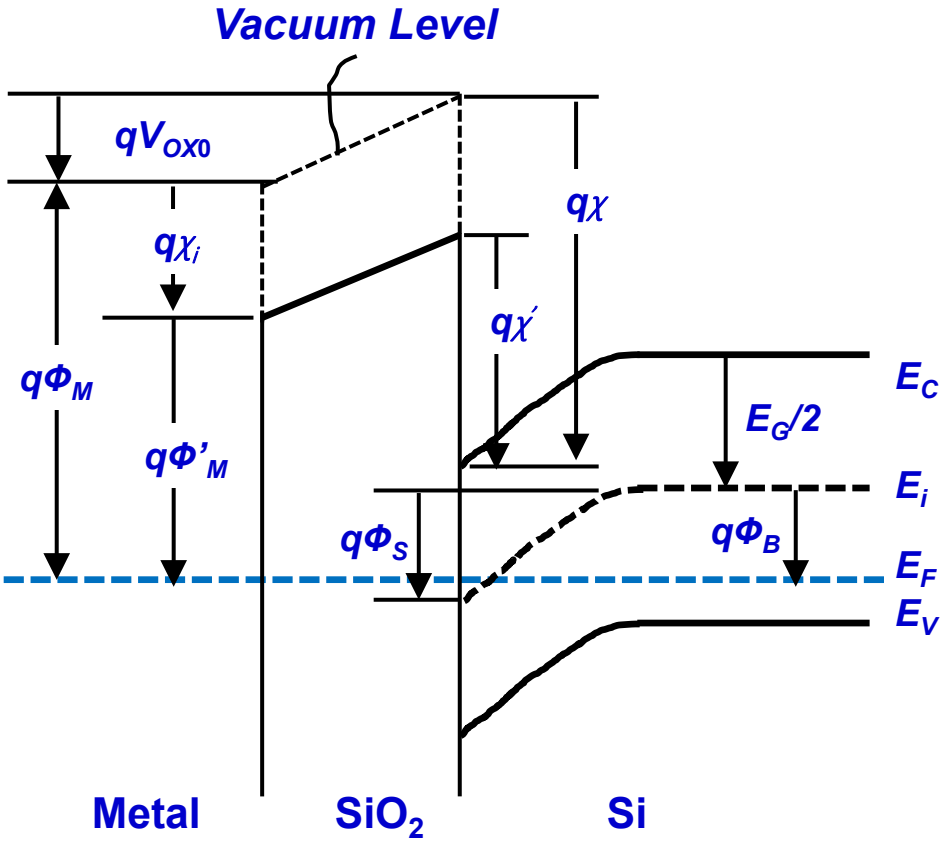


3. One-sided abrupt-junction approximation.

   at $X = W$, $p = N_A = p_{p0}$, and n = n_{p0}.

Flat band Condition for $\Phi_{MS} \neq 0$, $V_G = V_{FB}$





$V_{FB} = \Phi_M - \Phi_S = \Phi_{MS}$

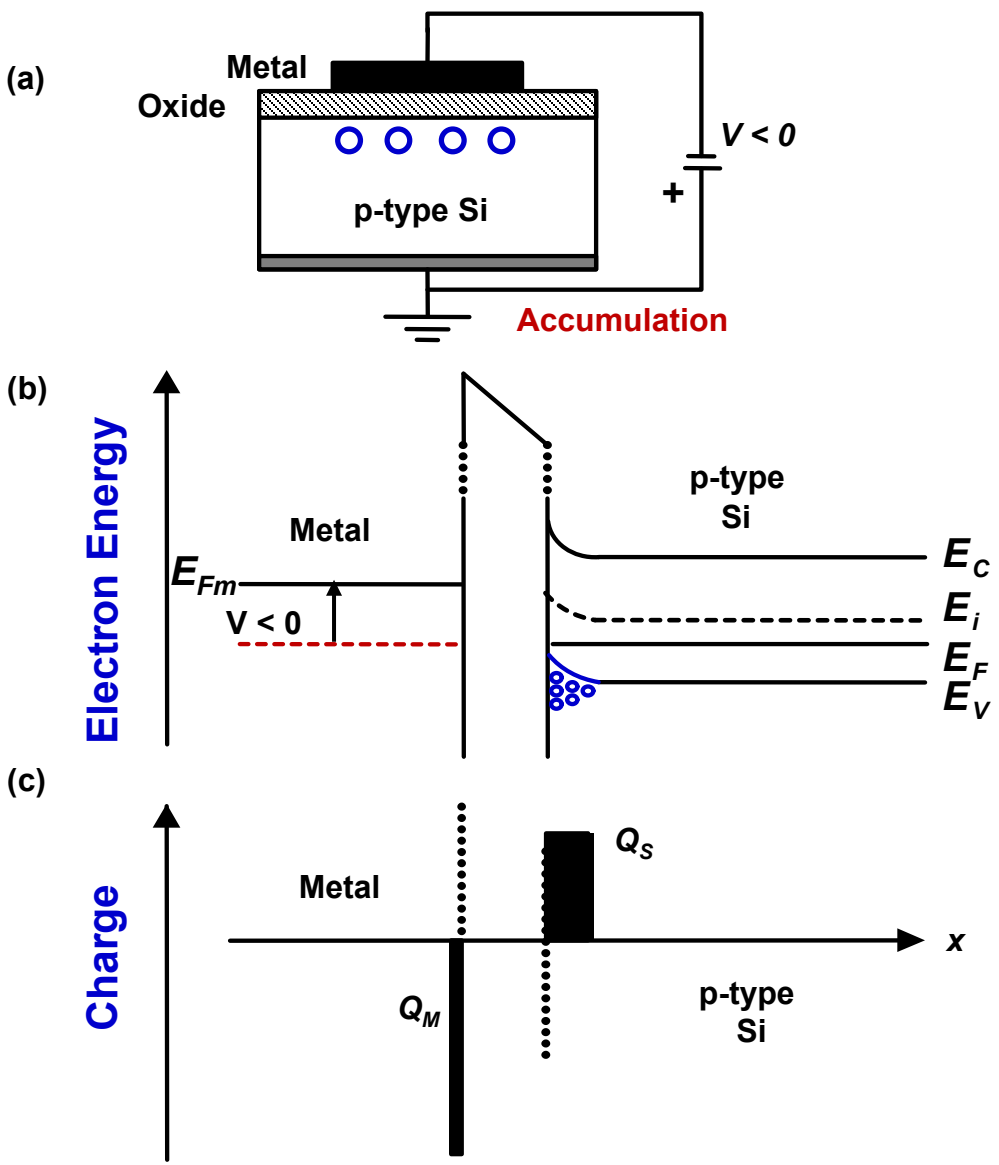
Considering fixed charges at the interface

$$V_{FB} = \Phi_{MS} - \frac{Q_f}{C_{ox}}$$

- Φ_S (surface potential): voltage drop across Si-substrate. total amount of band-bending downward

$$\Phi_S = \frac{E_{i,bulk} - E_{i,surface}}{q}$$

Accumulation ($V_G < V_{FB}$)



$$\Phi_s < 0$$

$$V_G = V_{FB} + V_{ox} + \Phi_s$$

$$\therefore V_G - V_{FB} = -\frac{Q_s}{C_{ox}} + \Phi_s$$

$$\rightarrow V_G - \Phi_{MS} = V_{ox} + \Phi_s$$

$$\Rightarrow Q_s \approx -C_{ox} \cdot (V_G - V_{FB})$$

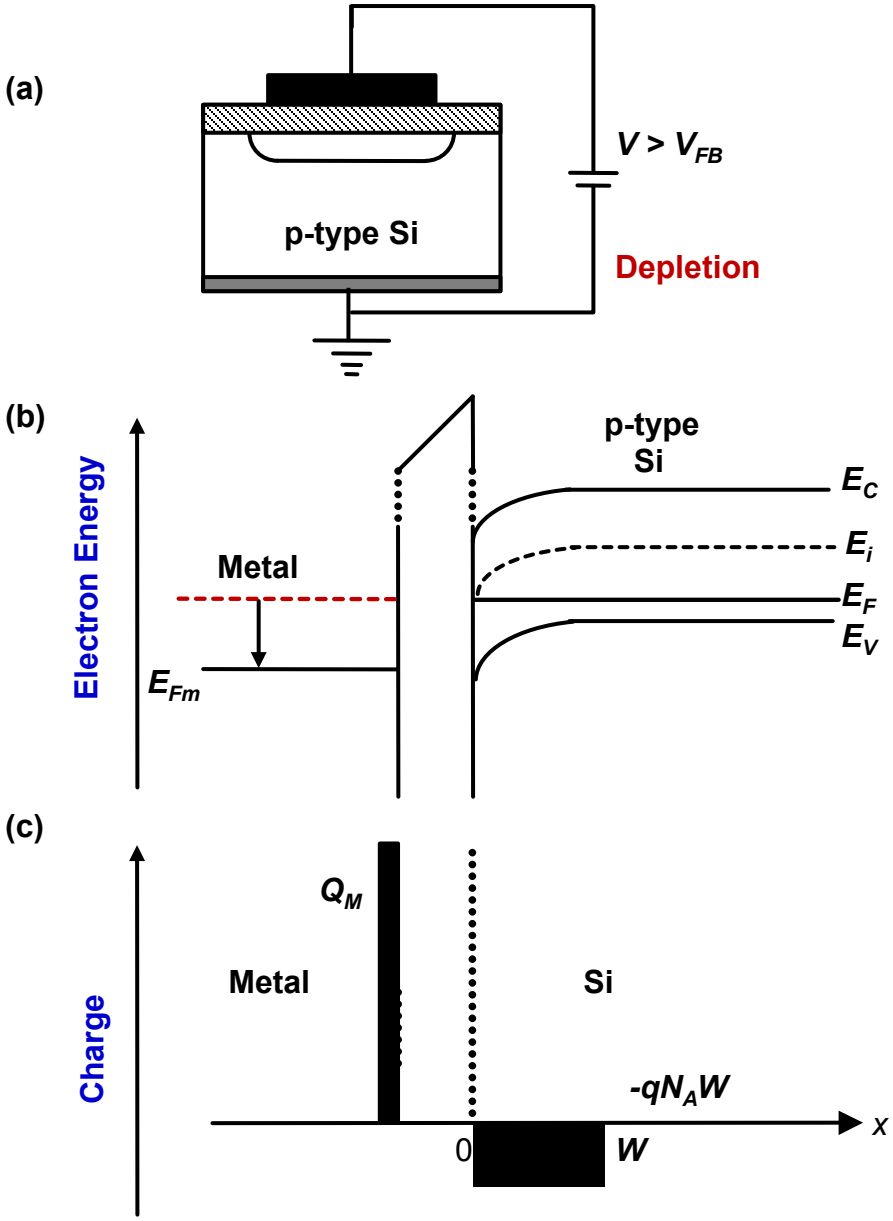
Φ_S is very small voltage drop.
Surface hole concentration:

$$p_s = N_A \cdot \exp\left(-\frac{q\Phi_s}{kT}\right)$$

Very small band bending can Introduce a large number of holes at the surface.

Most V_G across V_{OX}.

Depletion ($V_G > V_{FB}$)



Poisson's Equation $\dfrac{d^2\Phi}{dx^2} = -\dfrac{\rho(x)}{\varepsilon_{si}}$

$\rho(x) = -qN_A \Rightarrow \dfrac{d^2\Phi}{dx^2} = \dfrac{qN_A}{\varepsilon_{si}}$

$\dfrac{d\Phi}{dx} = \dfrac{qN_A}{\varepsilon_{si}}x + C_1$

Boundary Condition: $\dfrac{d\Phi}{dx} = 0$ at $x=W$

$C_1 = -\dfrac{qN_A}{\varepsilon_{si}}W$

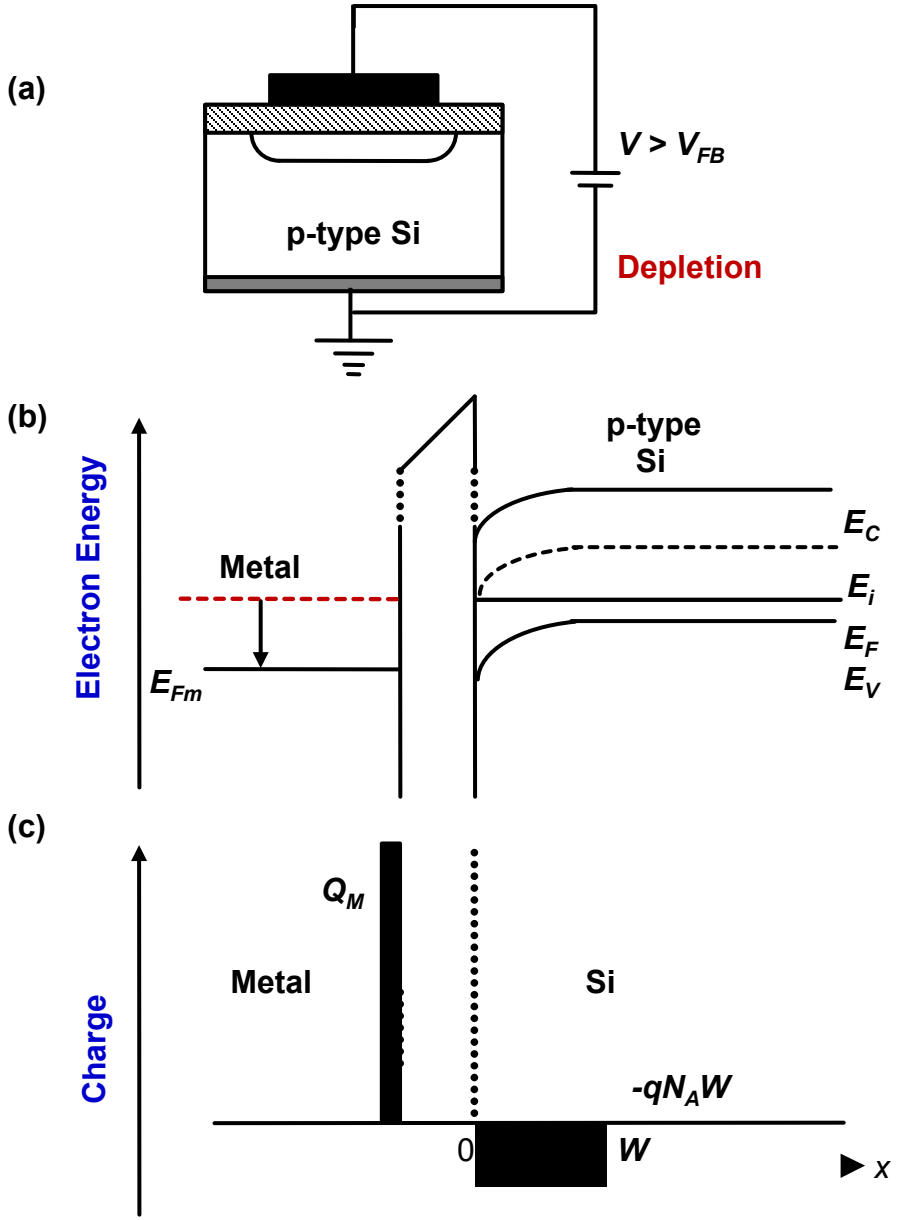
$\dfrac{d\Phi}{dx} = \dfrac{qN_A}{\varepsilon_{si}}(x-W) \Rightarrow \xi = \dfrac{qN_A}{\varepsilon_{si}}(x-W)$

$\Phi = \dfrac{1}{2}\dfrac{qN_A}{\varepsilon_{si}}(x-W)^2 + C_2$

Boundary Condition: $\Phi = 0$ at $x=W$

$C_2 = 0 \Rightarrow \Phi = \dfrac{1}{2}\dfrac{qN_A}{\varepsilon_{si}}(x-W)^2$

Depletion ($V_G > V_{FB}$)



Surface potential at $x = 0$

$$\Phi_S = \frac{1}{2}\frac{qN_A}{\varepsilon_{si}}W^2 \Rightarrow W = \sqrt{\frac{2\varepsilon_{Si}\Phi_S}{qN_A}}$$

Therefore,

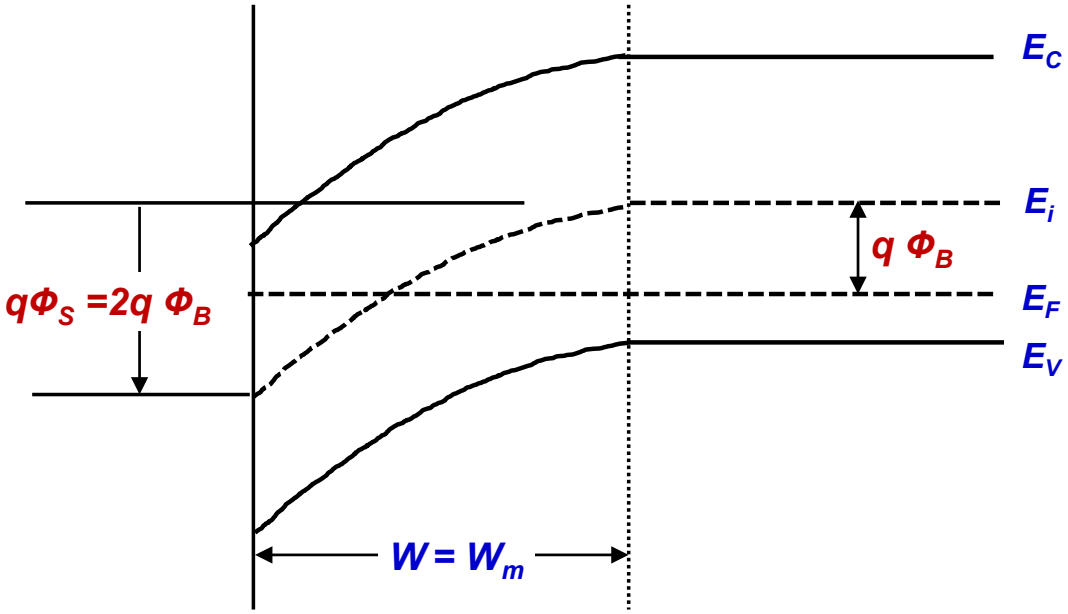
$$Q_S = -qN_AW = -\sqrt{2qN_A\varepsilon_{Si}\Phi_S}$$

We also have,

$$V_G - V_{FB} = -\frac{Q_S}{C_{ox}} + \Phi_S$$

$$V_G - V_{FB} = \frac{\sqrt{2qN_A\varepsilon_{Si}\Phi_S}}{C_{ox}} + \Phi_S$$

Threshold ($V_G = V_T$)



$$Q_S = Q_D + Q_N$$
$$\approx Q_D = -q \cdot N_A \cdot W$$

$$W_m \approx \sqrt{\frac{2\varepsilon_{Si}}{qN_A} \cdot (2\Phi_B)}$$

$$V_G - V_{FB} = \frac{\sqrt{2qN_A\varepsilon_{Si}(2\Phi_B)}}{C_{ox}} + 2\Phi_B$$

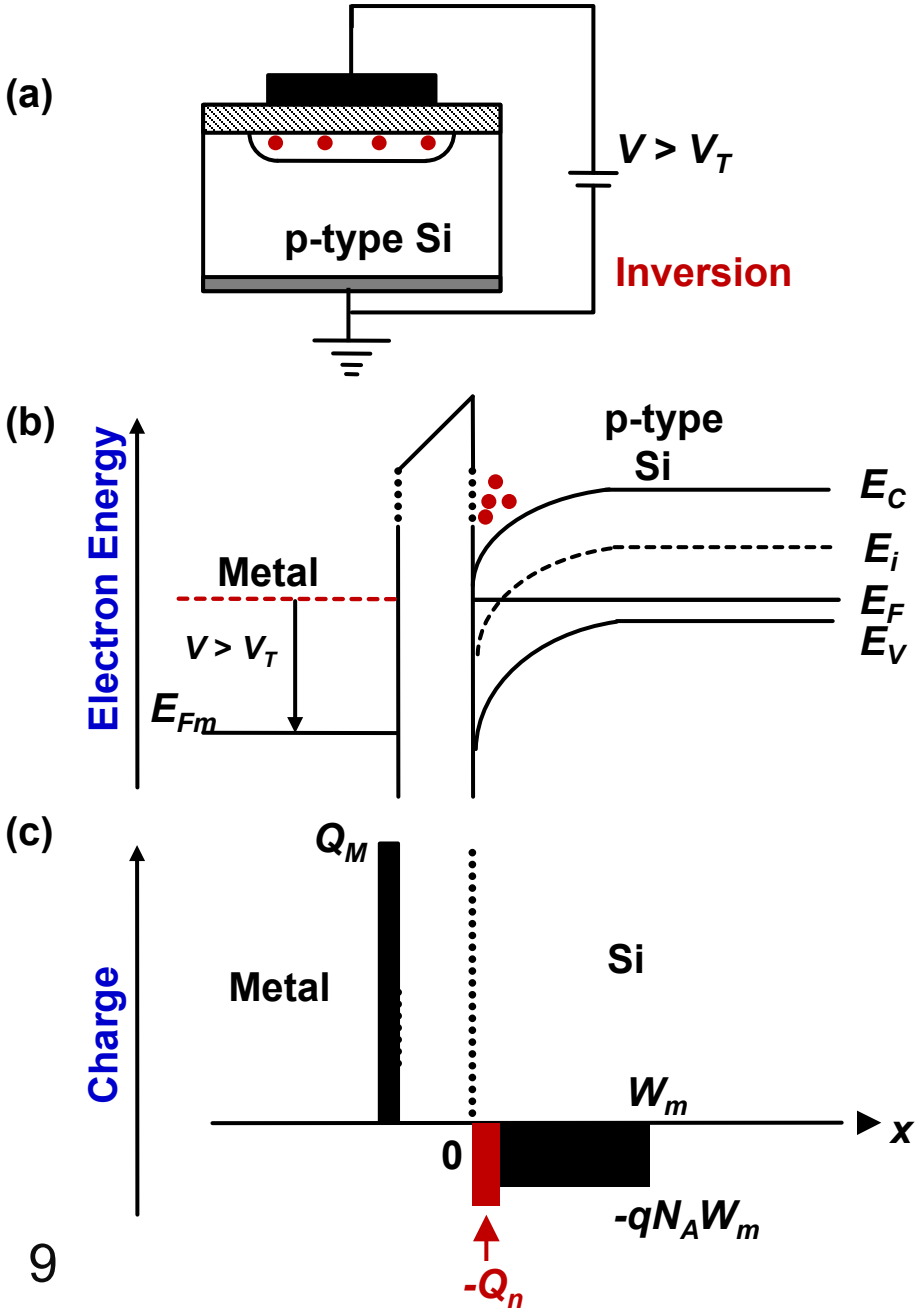
V_T is defined as the V_G where

$$n_S = p_{bulk} \quad \text{or} \quad \Phi_S = 2\Phi_B = 2\frac{kT}{q}\ln\left(\frac{N_A}{n_i}\right)$$

$$n = n_i \exp\left(\frac{E_F - E_i}{kT}\right), \quad p = n_i \exp\left(\frac{E_i - E_F}{kT}\right)$$

$$V_G = V_{FB} + \frac{\sqrt{2qN_A\varepsilon_{Si}(2\Phi_B)}}{C_{ox}} + 2\Phi_B = V_T$$

Strong Inversion ($V_G > V_T$)



At inversion,

$$\Phi_S > 2\Phi_B$$

We have

$$Q_S = Q_D + Q_n$$

$$V_G - V_{FB} = -\frac{Q_n}{C_{ox}} - \frac{Q_D}{C_{ox}} + \Phi_S$$

Therefore

$$V_G = V_{FB} - \frac{Q_n}{C_{ox}} + \frac{\sqrt{2qN_A\varepsilon_{Si}(2\Phi_B)}}{C_{ox}} + \Phi_S$$

$$= -\frac{Q_n}{C_{ox}} + V_{FB} + \frac{\sqrt{2qN_A\varepsilon_{Si}(2\Phi_B)}}{C_{ox}} + \Phi_S$$

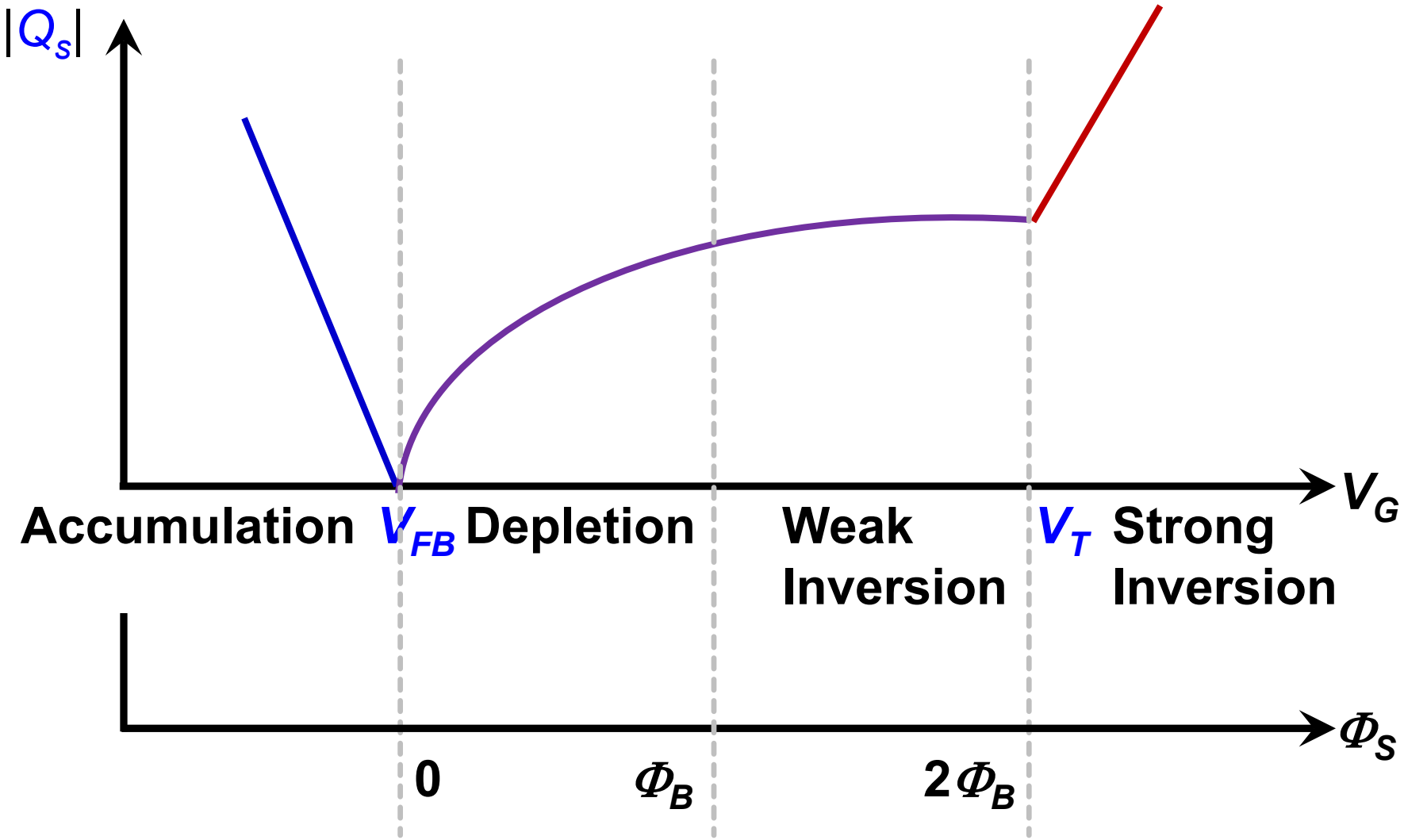
$$= -\frac{Q_n}{C_{ox}} + V_T$$

$$\Rightarrow \quad Q_n = -C_{ox}(V_G - V_T)$$

Summary for Depletion Approximation Analysis for P-substrate

Mode	Gate Voltage	Surface Potential	Substrate Charge
Accumulation	$V_G < V_{FB}$	$\Phi_s < 0$	$Q_s = -C_{ox} \cdot (V_G - V_{FB})$
Flat band	$V_G = V_{FB}$	$\Phi_s = 0$	$Q_s = 0$
Depletion	$V_G > V_{FB}$	$0 < \Phi_s < 2\ \Phi_B$	$Q_s = Q_D = -\sqrt{2qN_A\varepsilon_{si}\Phi_s}$
Threshold	$V_G = V_T$	$\Phi_s = 2\ \Phi_B$	$Q_s = Q_n + Q_D = -\sqrt{2qN_A\varepsilon_{si}(2\Phi_B)}$
Strong Inversion	$V_G > V_T$	$\Phi_s > 2\ \Phi_B$	$Q_s = Q_n + Q_D = -C_{ox} \cdot (V_g - V_T) - \sqrt{2qN_A\varepsilon_{si}(2\Phi_B)}$

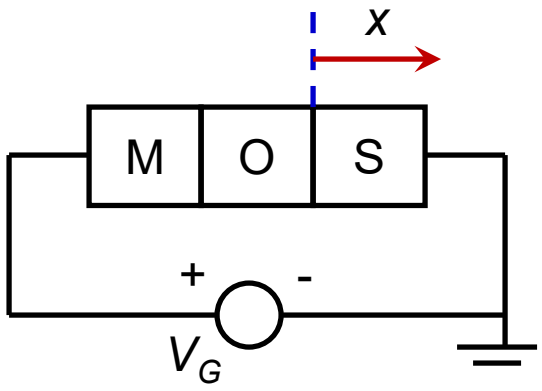
Q_S as a Function of V_G and Φ_s



Problems with Depletion Approximation

- Assumption of maximum depletion width W_m.
- Q_S in accumulation and weak inversion.
- Abrupt transition from one region to another region.

General Analysis



Define: $\Phi_S = \dfrac{E_{i,bulk} - E_{i,surface}}{q}$

$$n_p(x) = n_{p0}\exp\left(\frac{q\cdot\Phi(x)}{kT}\right) = n_{p0}\exp[\beta\cdot\Phi(x)]$$

$$p_p(x) = p_{p0}\exp\left(\frac{-q\cdot\Phi(x)}{kT}\right) = p_{p0}\exp[-\beta\cdot\Phi(x)]$$

$$\frac{d^2\Phi}{dx^2} = -\frac{\rho(x)}{\varepsilon_{si}}$$

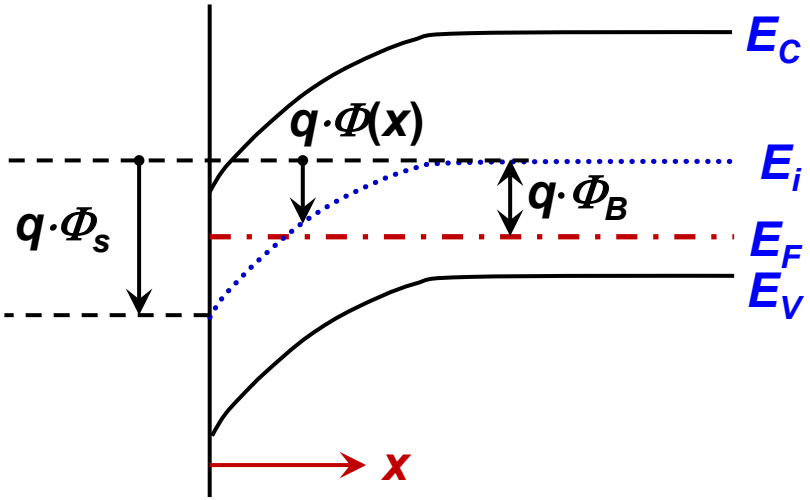
$$\rho(x) = q(N_D^+ - N_A^- + p_p(x) - n_p(x))$$

With charge neutrality in bulk,

$$\Phi = 0, \rho(x) = 0$$

$$N_D^+ - N_A^- = n_{p0} - p_{p0}$$

Oxide-Si Interface



Continued

General Analysis

Total charge density can be re-written as

$$\rho(x) = q[n_{p0} - p_{p0} + p_{p0}\exp(-\beta\Phi) - n_{p0}\exp(\beta\Phi)]$$
$$= q[p_{p0}(e^{-\beta\Phi} - 1) - n_{p0}(e^{\beta\Phi} - 1)]$$

Poisson's equation becomes

$$\frac{d^2\Phi}{dx^2} = -\frac{q}{\varepsilon_{si}}[p_{p0}(e^{-\beta\Phi} - 1) - n_{p0}(e^{\beta\Phi} - 1)]$$

Integrating this fro bulk toward surface

$$\int_0^{\frac{\partial\Phi}{\partial x}} (\frac{\partial\Phi}{\partial x})d(\frac{\partial\Phi}{\partial x}) = -\frac{q}{\varepsilon_{si}}\int_0^{\Phi}[p_{p0}(e^{-\beta\Phi'} - 1) - n_{p0}(e^{\beta\Phi'} - 1)]d\Phi$$

General Analysis Continued

Relation between electrical field $\xi(x) = -\frac{d\Phi}{dx}$

Potential Φ can be obtained by above integration

$$\varepsilon^2 = (\frac{2kT}{q})^2(\frac{qp_{p0}\beta}{2\varepsilon_{Si}})[(e^{-\beta\Phi} + \beta\Phi - 1) + \frac{n_{p0}}{p_{p0}}(e^{\beta\Phi} - \beta\Phi - 1)]$$

$$\varepsilon = \pm\frac{\sqrt{2}kT}{qL_D}F(\beta\Phi, \frac{n_{p0}}{p_{p0}})$$

Where $L_D = \sqrt{\frac{kT\varepsilon_{Si}}{p_{p0}q^2}} = \sqrt{\frac{\varepsilon_{Si}}{qp_{p0}\beta}}$

$$F(\beta\Phi, \frac{n_{p0}}{p_{p0}}) = \sqrt{(e^{-\beta\Phi} + \beta\Phi - 1) + \frac{n_{p0}}{p_{p0}}(e^{\beta\Phi} - \beta\Phi - 1)]}$$

Positive sign for Φ>0, and negative sign for Φ<0

Assuming $\Phi = \Phi_S$, electrical field at surface can be obtained

$$\varepsilon = \pm\frac{\sqrt{2}kT}{qL_D}F(\beta\Phi_S, \frac{n_{p0}}{p_{p0}})$$

$\Phi_S > 0$ if $\Phi > 0$

$\Phi_S < 0$ if $\Phi < 0$

General Analysis Continued

Space charge per unit area by Gauss's Law

$$Q_s = -\varepsilon_{si}\xi_s = \mp\frac{\sqrt{2}\varepsilon_{si}kT}{qL_D}F(\beta\Phi_s, \frac{n_{p0}}{p_{p0}})$$

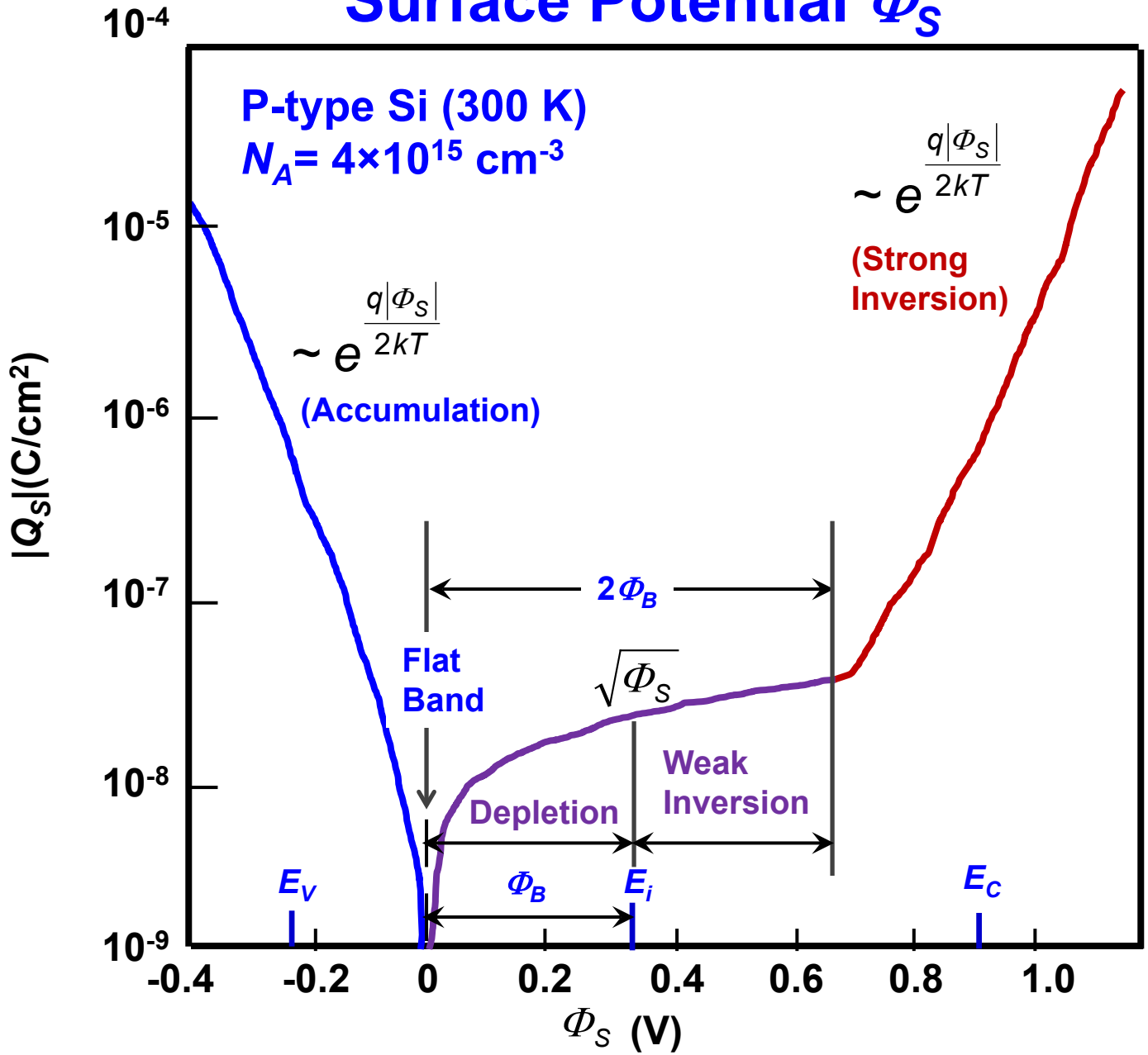
$$= \pm\sqrt{\frac{2q\varepsilon_{si}}{\beta}}\cdot\left[p_{p0}(e^{-\beta\Phi} + \beta\Phi - 1) + n_{p0}(e^{\beta\Phi} - \beta\Phi - 1)\right]^{\frac{1}{2}}$$

$$V_G - V_{FB} = -\frac{Q_s}{C_{ox}} + \Phi_s$$

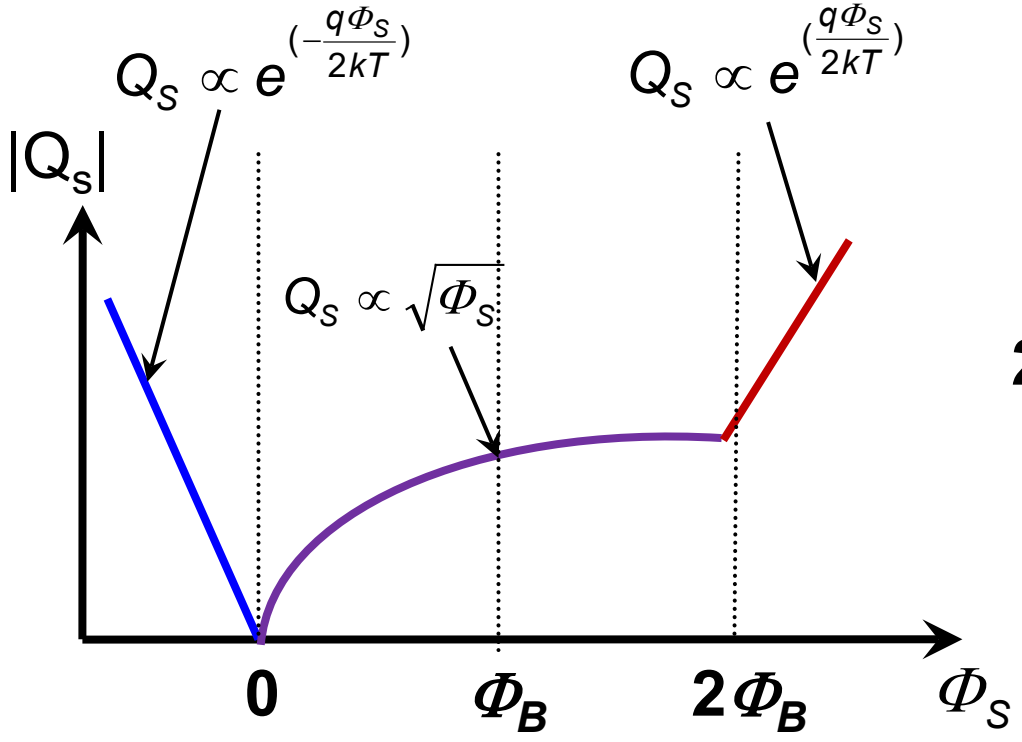
Q_S and Φ_S can be solved of V_G, N_A, and C_{OX}.

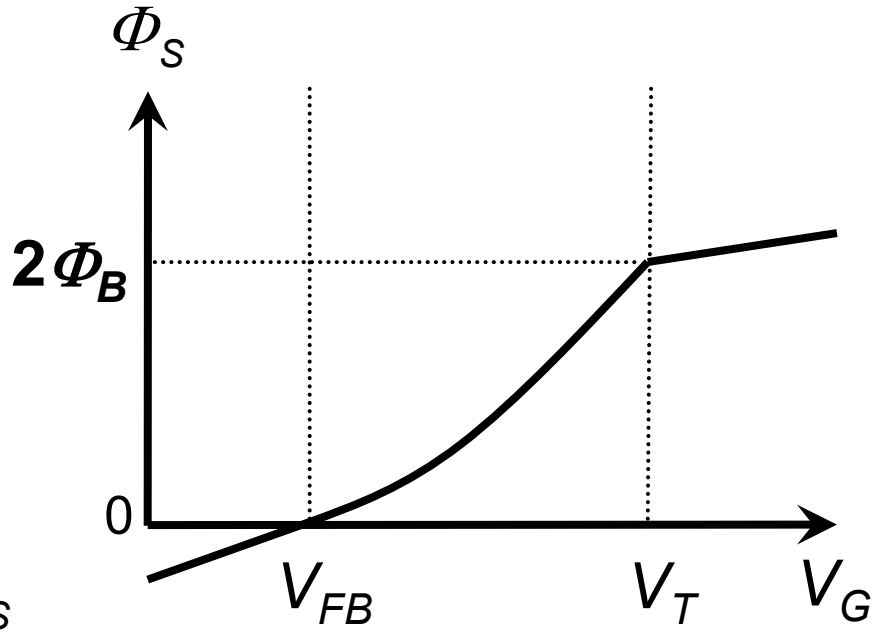
The expressions are valid for all regions of V_G.

Space Charge Density Q_S vs. Surface Potential Φ_S



Q_s vs. Φ_S and Φ_S vs. V_G





$$Q_s = \pm\sqrt{\frac{2q\varepsilon_{si}}{\beta}}\cdot\left[p_{p0}(e^{-\beta\Phi}+\beta\Phi-1)+n_{p0}(e^{\beta\Phi}-\beta\Phi-1)\right]^{\frac{1}{2}}$$

$$\frac{d\Phi}{dx}=\sqrt{\frac{2q}{\beta\cdot\varepsilon_{si}}}\cdot\left[F(\beta\Phi_s,\frac{n_{p0}}{p_{p0}})\right]^{\frac{1}{2}}$$

Space Charge Density Q_s

(a) $\Phi_S < 0$, $Q_s > 0$ (Accumulation)

$p_{po} \cdot e^{-\beta\phi}$ is dominant $\Rightarrow$ $Q_s \propto \exp\left(-\frac{\beta \cdot \phi_s}{2}\right)$

(b) $\Phi_S = 0$, $Q_S = 0$ (Flat band)

(c) $0 < \Phi_S < \Phi_B$, $Q_s < 0$ (Depletion)

$p_{p0} > n_{p0}$, $\beta\Phi_S >> e^{(-\beta\Phi_S)}$, $n_{p0} \cdot e^{\beta\Phi_S}$ is negligible

$\Rightarrow$ $Q_S \propto \sqrt{2q\varepsilon_{Si}N_A\Phi_S}$

(d) $\Phi_S > 2\,\Phi_B$, $Q_s < 0$ (Strong Inversion)

$n_{p0} \cdot e^{\beta\Phi}$ is dominant

$\Rightarrow$ $Q_s \propto \exp\left(\frac{\beta \cdot \phi_s}{2}\right)$